AI-driven Software Engineering

Guardrails for the Age of Vibe Programming

Dr. Stefan Trenkwalder
Code Praxis GmbH · PyCon Austria 2026

https://github.com/Code-Praxis/pycon-at-2026-workshop-ai-tdd

Dr Stefan Trenkwalder
Software Engineer

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Healthy Software is Engineered

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wooclap

AI is just amazing …

Low Autonomy ⟷ High Autonomy

Inline
Autocomplete

(Copilot)

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Inline Autocomplete	Chat Assistants
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(Copilot)	(ChatGPT)	(Cursor)	(Devin)

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Inline Autocomplete	Chat Assistants	AI-native IDEs	Coding Agents	Full-stack App Builders
(Copilot)	(ChatGPT)	(Cursor)	(Devin)	(Lovable)

Usage

// ADOPTION

- 92% of US developers use AI coding tools daily (Hashnode 2026)
- AI generates 46% of all new code
- 70% of engineers use 2–4 AI tools simultaneously

Usage without Trust

// ADOPTION

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// CALLOUT

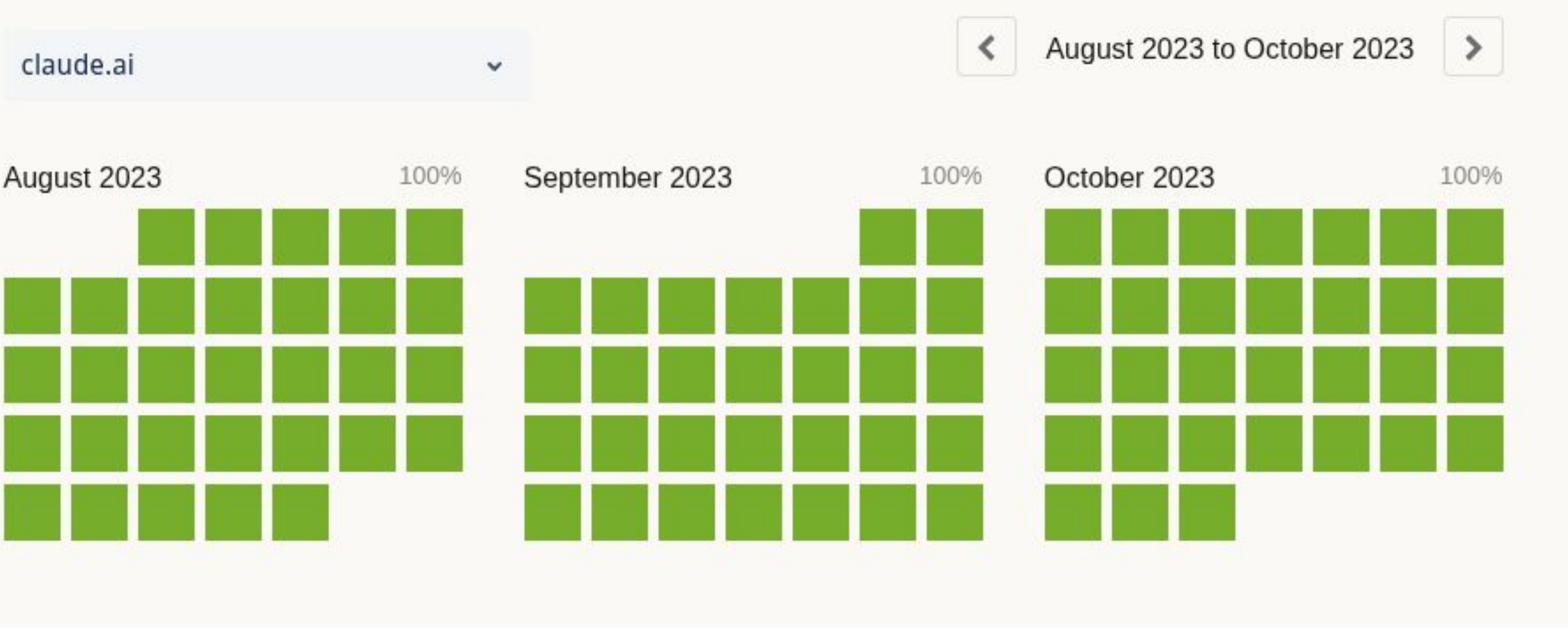
29 %

Trust in AI code accuracy

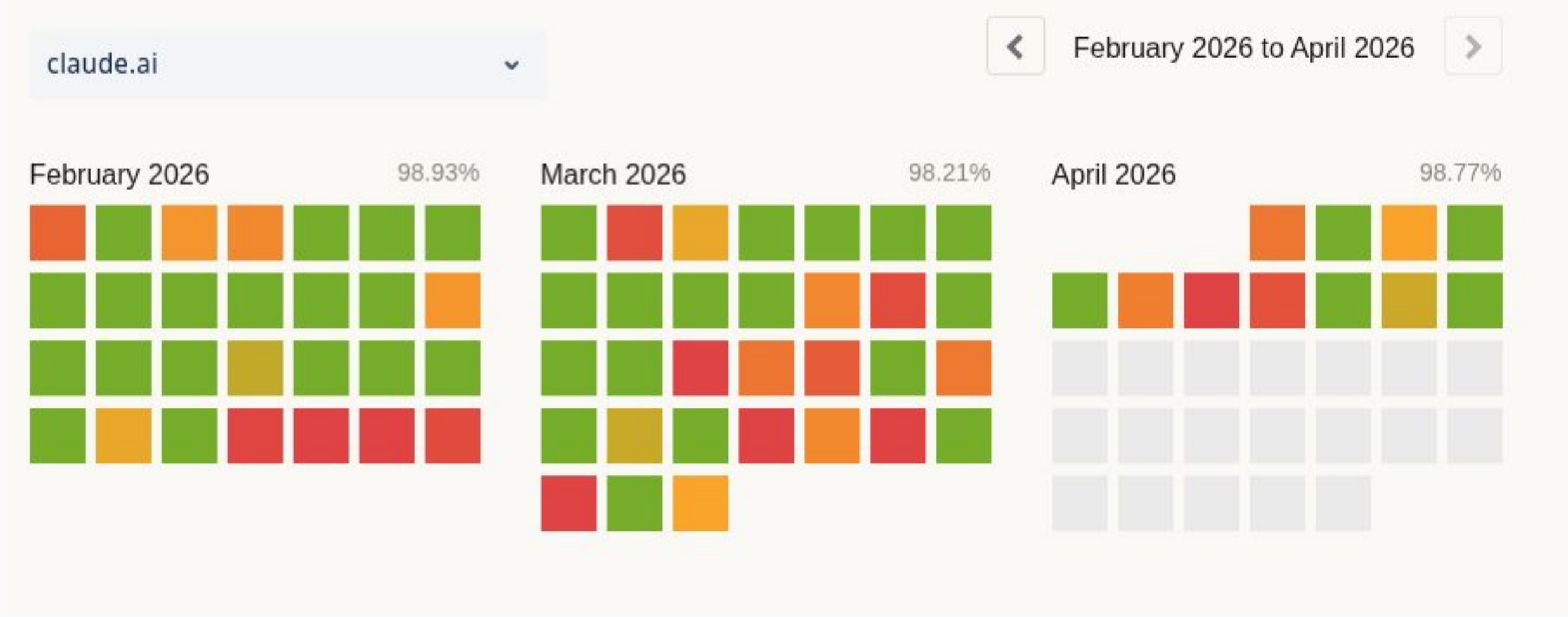
— Stack Overflow 2025

What is your biggest failure or frustration with AI-generated code?

Claude.ai


claude.ai
August 2023 to October 2023
August 2023
100%
September 2023
100%
October 2023
100%

Claude.ai


claude.ai
February 2026 to April 2026
February 2026
98.93%
March 2026
98.21%
April 2026
98.77%

Claude is not the only one …

- AWS outage (Oct 2025): Netflix was down too

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Claude is not the only one …

- AWS outage (Oct 2025): Netflix was down too
- GitHub: uptime dropped below 90%
- Stanford study (Perry/Boneh):
  - 88% of developers report negative AI impacts on code quality (Sonar 2026)
  - 53% note AI produces code that "looks correct but is unreliable"
  - 45% report debugging AI code takes more time than writing it manually
  - 80% of developers bypass security policies when using AI tools (Snyk)
  - Only 10% scan AI-generated code before shipping

Where do we go from here?

Where do we go from here?

Guard Rails

We know this problem .. We dealt with this for decades

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Industry

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No Influence

Industry

Guardrails: ...

01

Formatter enforcing

For consistent code

Black

Ruff

Guardrails: ...

01

Formatter enforcing
For consistent code

Black

Ruff

02

Linter, Security Scanners, (Reviewing) enforcing
For consistent typing, coding standards, security standards, ...

Ruff

mypy

Sonar

CodeRabbit

Guardrails: ...

01

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02

Linter, Security Scanners, (Reviewing) enforcing
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03

Test Suite
For correct behaviour

Guardrails: ...

01

Formatter enforcing
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Linter, Security Scanners, (Reviewing) enforcing
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03

Test Suite
For correct behaviour

We need a Workflow that guarantees tests

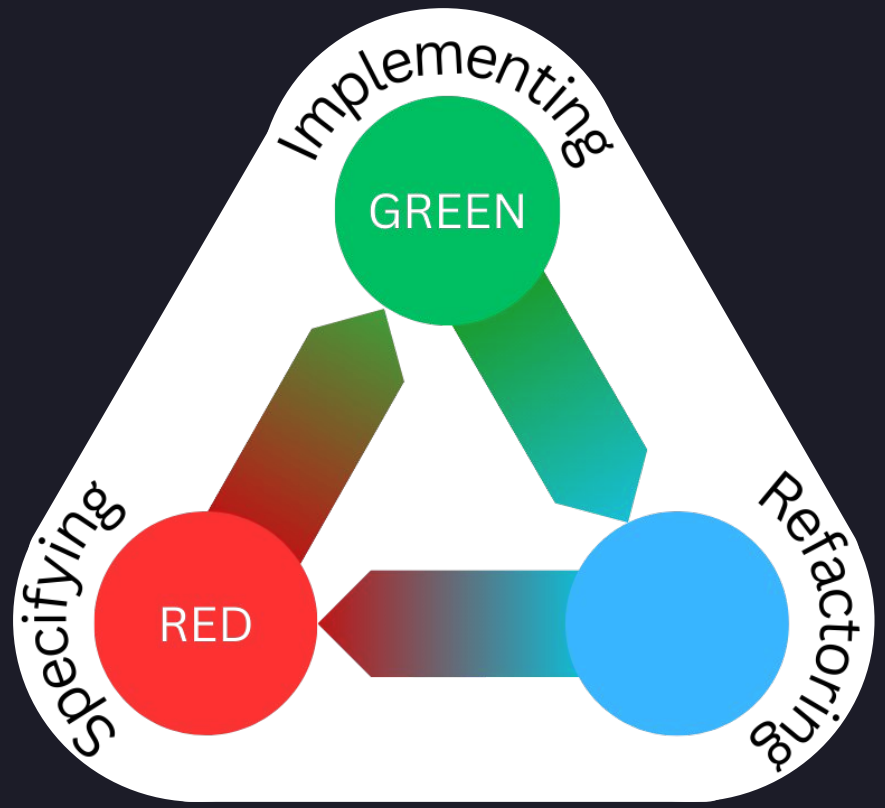
A Workflow that guarantees tests

Test-Driven Development

- Write a failing test
  (specify a desired behaviour)

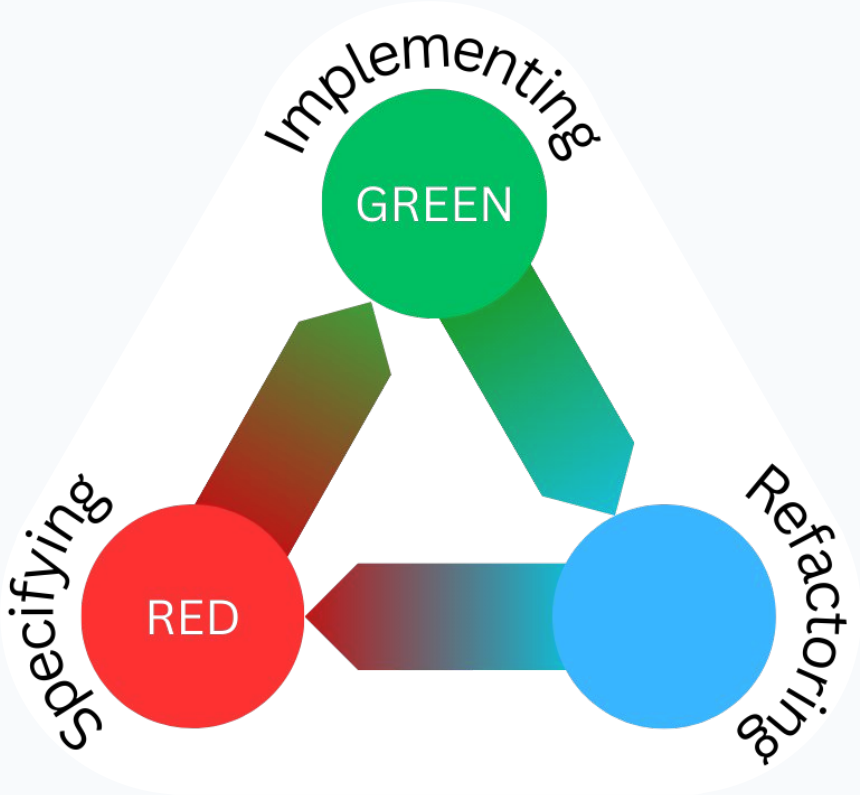
- Write production code to make it pass
  (implement the behaviour)

- Refactor the solution
  (clean the code up, look at design)



Laws of Test-Driven Development (TDD)

1. Only write production code to make a failing test pass
2. Don’t write more test code than is sufficient to fail
3. Don’t write more production code than is sufficient to pass a failing test



TDD Side Effects

6-35% slower
40-80% fewer defects

Jeffries, Ron, and Grigori Melnik. "Guest Editors' Introduction: TDD--The Art of Fearless Programming." IEEE Software 24.3 (2007): 24-30.

Family of studies	Type	Development time analyzed	Legacy project?	Organization studied	Software built	Software size	No. of participants	Language	Productivity effect	Quality effect
Sanchez et al.[6]	Case study	5 years	Yes	IBM	Point-of-sale device driver	Medium	9–17	Java	Increased effort 19%	40%†
Bhat and Nagappan[7]	Case study	4 months	No	Microsoft	Windows networking common library	Small	6	C/C++	Increased effort 25–35%	62%†
	Case study	≈7 months	No	Microsoft	MSN Web services	Medium	5–8	C++/C#	Increased effort 15%	76%†
Canfora et al.[8]	Controlled experiment	5 hours	No	Soluziona Software Factory	Text analyzer	Very small	28	Java	Increased effort by 65%	Inconclusive based on quality of test
Damm and Lundberg[9]	Multi-case study	1–1.5 years	Yes	Ericsson	Components for a mobile network operator application	Medium	100	C++/Java	Total project cost increased by 5–6%	5–30% decrease in fault-slip-through rate; 55% decrease in avoidable fault costs
Melis et al.[10]	Simulation	49 days (simulated)	No	Calibrated using Klondike-Team and Quinary data	Market information project	Medium	4‡	Smalltalk	Increased effort 17%	36% reduction in residual defect density
Mann[11]	Case study	8 months	Yes	PetroSleuth	Windows-based oil and gas project management with statistical modeling elements	Medium	4–7	C#	n/a	81%§; customer and developers' perception of improved quality

TDD Side Effects

You naturally build up a comprehensive test suite!

https://github.com/Code-Praxis/pycon-at-2026-workshop-ai-tdd

Exercise: Shipping Costs

The application must calculate shipping costs based on the following rules:

1. **Base Rate:** 5.00 € flat fee for any shipment.
2. **Weight Tiers**:
   - 0 to 5 kg: 1.00 € per kg
   - Over 5 kg to 20 kg: 1.50 € per kg
   - Over 20 kg: 2.00 € per kg
3. **Distance Zones:**
   - Zone A (Local): 1x multiplier
   - Zone B (Domestic): 1.5x multiplier
   - Zone C (International): 2.5x multiplier
4. **Express Delivery:** Adds a flat 15.00 € surcharge *after* all other calculations.
5. **Edge Cases:**
   - Negative weight or distance should raise an `ValueError`.
   - Maximum weight is 100 kg; exceeding this raises a `ValueError`.

Manual TDD Exercise

— To understand the workflow

Retrospective with focus on the Workflow

Cons

- Slow(er)
- Double the effort (twice as much to maintain)
- Rigid rigour
- Different to how programming is taught / learned

Pros

- Small iterations/improvements
- Quick feedback loop (on correctness and design)
- Behaviour specifications are tested repeatedly
- Fixed step to improve the code/design/architecture
- Different roles for each step

Prompt Only

— To get started with AI

Retrospective on Prompting Only

- Review the code
  - Review is more difficult than writing -> LGTM
  - Cognitive Debt : Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task - Kosmyna et al. (Artificial Intelligence) - https://arxiv.org/abs/2506.08872
  - Comprehension Debt:
    - https://margaretstorey.com/blog/2026/02/09/cognitive-debt/
    - https://martinfowler.com/fragments/2026-02-13.html
  - How to Kill the Code Review - https://www.latent.space/p/reviews-dead

Retrospective on Prompting Only

- The Anthropic RCT (2025/2026) - https://www.anthropic.com/research/AI-assistance-coding-skills
  - 52 junior engineers learning a new library
  - Manual coding group: 67%
  - AI-assisted group: 50%
  - Full delegation to AI: Below 40% (effectively failing)

  - The largest gap appeared in debugging and skill to validate AI code

Retrospective on Prompting Only

- METR RCT (July 2025) - https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/
  - 16 experienced open-source developers, 246 real issues, their own codebases:
  - Predicted speed gain (before study): +24%
  - Perceived speed gain (after study): +20%

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  - 16 experienced open-source developers, 246 real issues, their own codebases:
  - Predicted speed gain (before study): +24%
  - Perceived speed gain (after study): +20%
  - **Actual measured speed: -19% (slower)**

Retrospective on Prompted TDD

- The Stanford study (Perry/Boneh) - https://ee.stanford.edu/dan-boneh-and-team-find-relying-ai-more-likely-make-your-code-buggier

Developers using AI write significantly **less secure code** and are more likely to **believe it is secure**.

Prompted TDD

— Guardrail your AI

Retrospective on TDD with AI

- **TDFlow: Agentic Workflows for Test Driven Development** - Han et al. (EACL 2026) - https://arxiv.org/abs/2510.23761
  - 88.8% pass rate on SWE-Bench Lite (an absolute improvement of 27.8% over the next best system)
  - Best results when tests have been witten by developers.

- **Enhancing LLM Code Generation Capabilities through Test-Driven Development and Code Interpreter** - Jalil et al. (AACL-IJCNLP 2025) - https://arxiv.org/abs/2511.12823
  - Small open-source models with TDD and CI (feedback) outperformed expensive/commerical models.

Retrospective on Prompted TDD

- Repetitive
- Copy paste promts
- Really depends on the prompt
- Review the code after each step
  - Review is more difficult than writing
  - Most just approve (LGTM)

Agentic TDD Exercise

— High-throughput coding

Thank You.

Healthy Software is Engineered.

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